Decision Tree Complexity: Investigating Overfitting through Tree Depth Using the Titanic Dataset

Github: https://github.com/bijjivinay40-study/Machine-Learning-1/tree/main

1. Overview:
Machine learning models must balance simplicity and predictive performance. If a model is too simple, patterns will be present in the data that is not detected. On the other hand, high complexity of a model results in learning noise in the training data instead of patterns. Underfitting and overfitting are the names of these two problems.

Decision trees are common machine learning algorithms that nicely illustrate this trade-off. A decision tree structure can be very simple with a handful of decision rules. It can also be a very complex structure that attempts to capture all the intricate details in the data set. Tree depth is one of the main factors controlling a decision tree's complexity, that is, the number of levels of decision splits it may create.

The depth of a tree controls how much the model learns from an input. The shallow tree has little or no depth and may not be able to capture important relations effectively. A very deep tree, on the other hand, may overfit the training data resulting in poor accuracy on unseen data.

This lesson examines how different tree depths affect model behaviour and model performance. Using Titanic Dataset, the experiment shows increasing tree depth and its effect on model complexity, training accuracy and generalization performance.

The tutorial explains how you can choose optimally complex models through decision tree visualizations, accuracy curves, and feature importance which through tree depth can cause either underfitting or overfitting.

2. Decision Trees as Classification Models
Decision Trees are supervised learning algorithms used for both classification and regression tasks. In classification problems, the model learns a sequence of decision rules that divide the dataset into groups with similar target values.

A decision tree is composed of three main elements:

- **Root node** – the starting point of the decision process
- **Decision nodes** – intermediate points where the dataset is split based on feature values
- **Leaf nodes** – final nodes representing classification outcomes

Every internal node in the tree represents a decision rule that is based on one input feature. According to this rule, data is divided into smaller data sets. The process continues until a stopping condition occurs, for example reaching a specified tree depth.

Gini impurity or entropy is applied by Decision Trees to decide which feature to use for splitting the data. Measures assess the discriminatory power of a feature in separating observations.

Decision trees are highly interpretable, a major advantage of decision trees. The order in which splits occur gives good insight into how predictions are reached.

3. Understanding Tree Depth

The depth of a tree is defined as the number of levels of the decision tree from the root node to the farthest leaf node.

The depth of the tree decides how many sequential decisions can be made by the model on the observation.

A model with few decision rules will be a simple model and will have a small tree depth. These models are more interpretable, but they may miss out on capturing complex patterns in the data.

The model can create many decision rules with a large tree depth and capture complex relationships well. Yet, this incremental complexity might lead the model to memorize the training data rather than learning general patterns.

This behaviour closely resembles the concept of "overfitting." In overfitting, the model does a good job at the training set, but terrible at others.

4. Dataset Description

The dataset used in this tutorial is the **Titanic dataset**, which contains information about passengers aboard the RMS Titanic. The objective of the classification task is to predict whether a passenger survived the disaster based on personal and travel-related attributes.

The dataset includes several features describing passengers, such as:

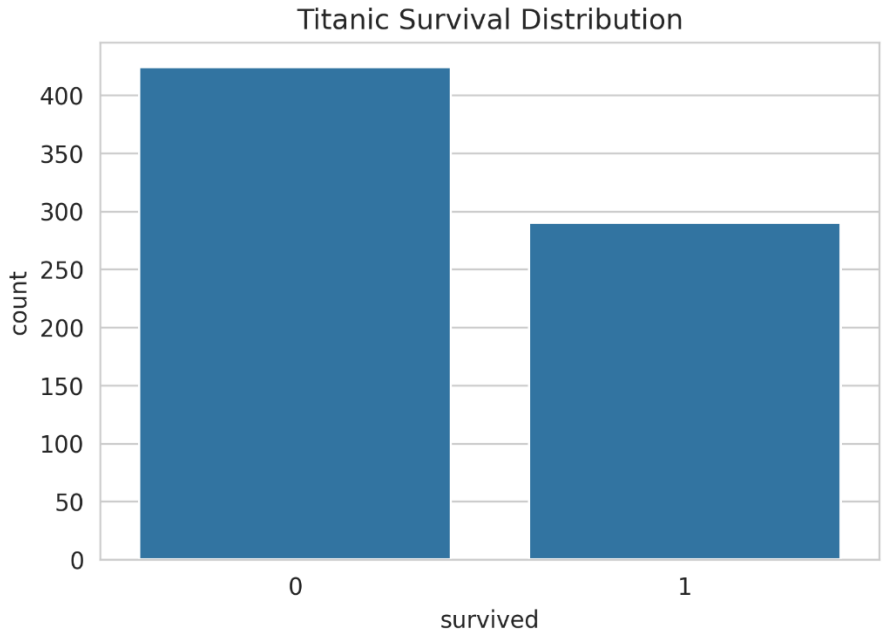
- passenger class (Pclass)
- gender (Sex)
- age (Age)
- ticket fare (Fare)

The target variable is:

- **Survived** – indicating whether the passenger survived (1) or did not survive (0).

5. Initial Data Exploration

Before building the decision tree models, it is useful to examine the distribution of the target variable.



The visualization shows the number of passengers who survived compared with those who did not survive. The dataset indicates that a larger proportion of passengers did not survive, highlighting the challenging nature of the classification task.

6. Experimental Setup

To investigate how tree depth influences model performance, several decision tree models were trained with different maximum depth values.

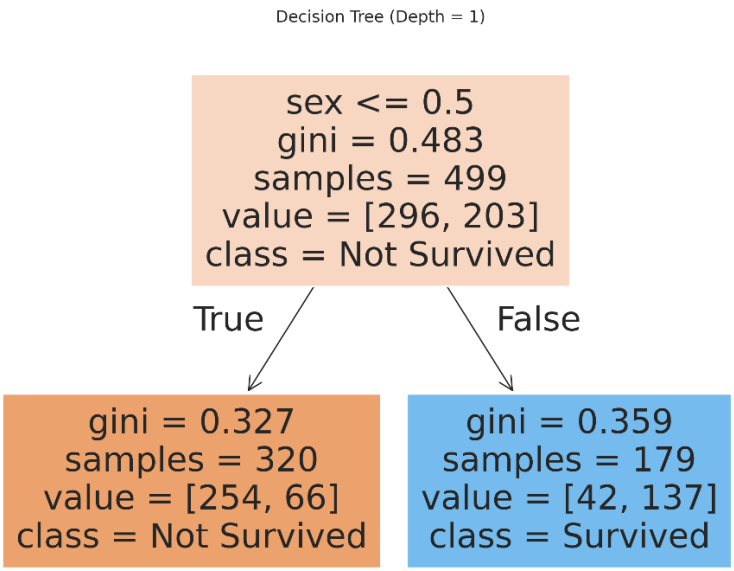
The dataset was first divided into **training** and **testing** sets. The training data was used to train the decision tree models, while the testing data was used to evaluate model performance.

Three tree depth configurations were tested:

- Depth = 1 (very shallow tree)
- Depth = 3 (moderate complexity)
- Depth = 10 (very deep tree)

By comparing the performance of these models, it is possible to observe how increasing tree depth affects prediction accuracy and model complexity.

→Shallow Decision Tree (Depth = 1)

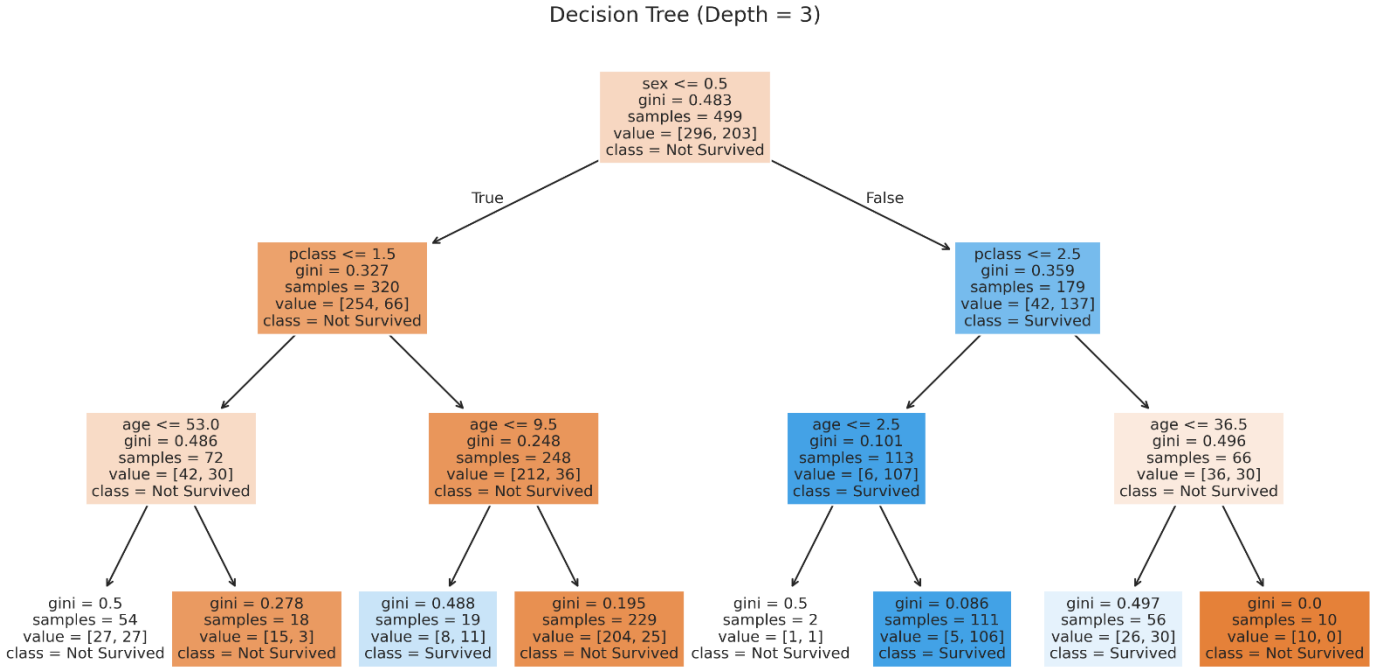


The model can make just one split on the dataset if we restrict the maximum depth of the tree to 1. This generates a basic model which has limited predictive power.

When a model is shallow, it may not capture meaningful relationships in the features and target variable. Accordingly, this will result in the model being underfitted, thus, cannot represent the underlying structure of data.

The model is simple and interpretable; however, the predicted accuracy typically is lower than with more complex trees.

→Medium Complexity Tree (Depth = 3)

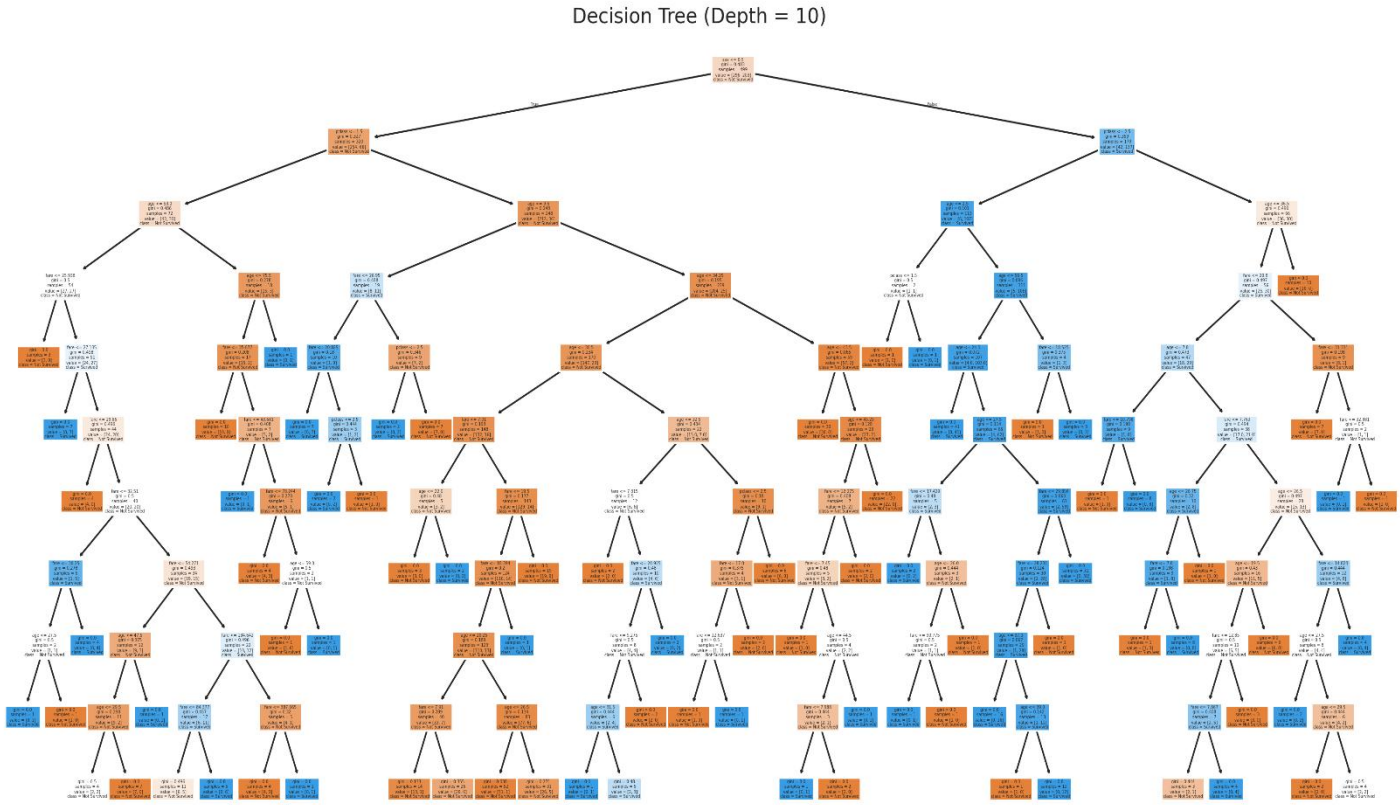


When the tree gets deeper, the model can make more splits which means it captures more complex relationship between features and survival.

Due to its depth of three levels, the model can merge several features for prediction. This usually leads to a more balanced approach to learning from data.

Typically, trees of moderate sizes possess good generalization performances.

→Deep Decision Tree (Depth = 10)



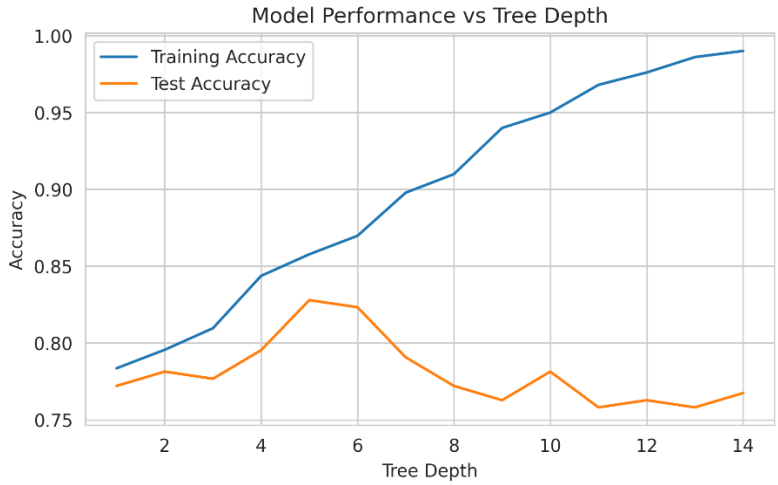
Increasing the maximum depth at a significant level makes the decision tree too complex. The model keeps dividing the data until it learns very specific aspects of the training data.

Although this can lead to better training accuracy, it often results in overfitting. The model starts to identify noise, and irregularities present in the training data rather than general patterns.

Models that are overfitted do not do well on unseen data.

7. Model Performance Analysis

To understand how tree depth affects predictive performance, model accuracy was measured for different tree depths.



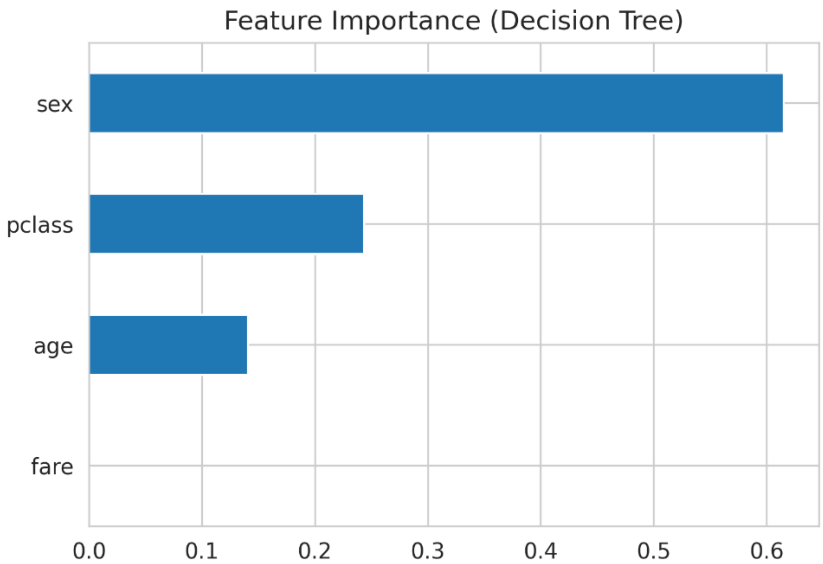
The graph illustrates how training accuracy increases as tree depth grows. However, testing accuracy does not always improve at the same rate.

Initially, increasing depth improves model performance because the model becomes capable of capturing meaningful patterns in the dataset.

After a certain point, however, testing accuracy begins to decline even though training accuracy continues to increase. This behaviour indicates overfitting.

8. Feature Importance

Decision Trees also provide information about the relative importance of each feature in making predictions.



- The analysis of variable importance indicates which variables contributed most strongly to predicting survival.
- Analyses of Titanic often find gender to be one of the strongest features as survival rates of women and men differ significantly.
- Understanding which features are important can help us make sense of what our model is doing.

9. Discussion

The findings of this experiment show how the depth of a decision tree affects model complexity and predictions. If the tree depth is made small then the resulting model will contain a small number of decision rules. This makes the model easy to interpret, but it often overlooks important relationships in the dataset. Underfitting occurs when the model is too simple to capture the underlying trends of the data.

With increasing tree depth, the model's predictions grow increasingly accurate as it identifies important relationships and interactions among more features. Because we used these techniques, we could use more variables like passenger class, age, gender, etc. in making predictions which improved accuracy. At intermediate depths, the decision tree attained the perfect balance between simplicity and predictive ability.

However, when tree depth was very large, the model overfits the training data. When the model learns particular principles that provides unique information on training data but do not propagate to new data, the model is overfitted.
This occurs in the accuracy versus depth plot, where training accuracy keeps increasing and testing accuracy does not continue to improve or stops improving eventually.

Another important observation from this analysis is that of 'feature importance'. The decision tree model identifies which variable matter most for prediction purpose. The gender and passenger classes in the Titanic dataset are usually influential for determining the chances of survival.

In conclusion, the experiment shows how critical it is to control model complexity in decision trees.

10. Limitations

This experiment used a limited number of features from the Titanic dataset. Incorporating additional variables may improve classification performance.

Furthermore, only a few tree depth values were tested. A more systematic evaluation could involve testing a larger range of depth values.

10. Future Work

In the future work, cross-validation can be done to obtain reliable estimates of the performance. Cross-validation is a technique for evaluating a model by fitting it to several places of a dataset.

One further option could be to examine the possibilities of other decision tree complexity controlling methods such as minimum sample per leaf or pruning.

It may also be insightful to compare it with other machine learning algorithms such as Random Forests or Gradient Boosting models.

11. Conclusion

This tutorial examined how the complexity of decision trees affects model behaviour by analyzing how depth affects classification performance.

We see that shallow trees underfit the data and very deep trees may overfit the training data. A tree depth between 4 and 6 is usually ideal for good results.

To design reliable predictive models, it is important to understand the effect of model complexity on performance.

Decision Trees offer a clear and intuitive framework for studying these concepts which makes them a useful resource for both machine learners and educators.

12. Reference

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